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Defined Poly(borosiloxane) as an Artificial Solid Electrolyte Interphase Layer for Thin-Film Silicon Anodes

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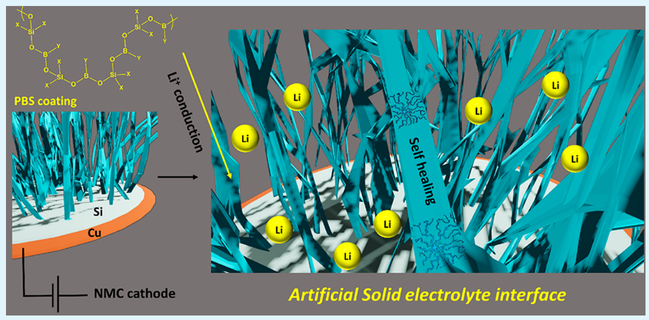
ABSTRACT: Solid electrolyte interphase (SEI) formation in Li-ion batteries is essential for good long-term performance of the cell. However, for electrodes exhibiting high volume expansion (like Si and Sn), continuous SEI formation can not only deplete electrolyte content but also increase cell impedance and hence mediocre performance. This is particularly detrimental in the case of thin-film electrodes where there is a minute amount of active materials loading. In the current work, defined poly(borosiloxane) (PBS) as an artificial polymeric SEI having self-healing, anion-trapping properties, and electron-deficient boron moiety is investigated. Studies on thin-film Si electrodes show excellent enhancement with polymeric coating in terms of cycling stability. These improvements are attributed to a combination of factors including the anion-trapping effect of tricoordinate boron, self-healing ability of PBS, and adherence to the electrode surface.



KEYWORDS: *Li ion batteries, Si anodes, Artifical SEI, polymeric coating, self-healing, anion trapping, cycling stability*

1. INTRODUCTION

Li-ion batteries have gained considerable interest owing to their impressive properties and low maintenance. Rapid commercial demand and graduation to large-scale applications have fueled battery research to the next level, where the focus has been to obtain greater capacity and higher voltage limits from the same footprint area. In this regard, silicon anodes have been the most well studied for electrodes because of their extraordinarily high capacity (∼3579 mA h g^{-1}), which is almost 10 times higher than the conventional graphite-based anodes. However, apart from the major issue of volume expansion (∼300 vol %), there are other serious concerns, like high irreversible capacity, continuous solid electrolyte formation due to a freshly exposed Si surface from volume expansion, low intrinsic conductivity, and aggregation, resulting in limited active site utilization by Li^+ ions.[1] In this regard, efforts have been directed individually toward each of the aforementioned problems, like carbon coating for increasing the conductivity,[2,3] excellent binders that curb volume expansion,[4−6] artificial solid electrolyte interphase to mitigate a rise in interfacial impedance,[7−9] prelithiation strategies to overcome the issue of irreversible electrolyte degradation,[10] etc. Poly(borosiloxane) (PBS)-based binders have been reported previously by Odom and co-workers[11] for performance enhancement in Si anodes because of their ability to adhere well onto the Si/SiO_2 surface via dipole−dipole interaction, their self-healing nature,[12,13] and the single-ion conductivity of tricoordinate boron. However, like most of the works on poly(borosiloxane)s, their work was also based on the utilization of a cross-linked framework, thereby not providing any clear mechanistic origin of performance evaluation. There have been many reports on the application of cross-linked PBS-based materials in different areas such as flexible conductors,[13] ion sensors,[14] and encapsulation of LEDs. However, there are only a few countable reports on well-defined PBS-based polymers with a defined mechanistic origin for their applications. We have previously reported a linear, highly alternating, and well-defined PBS polymer with tuned properties for ultrasensitive fluoride ion sensing[14] and have also elucidated their self-healing nature for application in corrosion protection of TiO_2 chips.[15] Our defined PBS materials with excellent self-healing properties at ambient temperature and Lewis acidic boron moiety can thus be an ideal candidates for Si anodes. To evaluate their contribution in mitigation of volume expansion and formation of a robust interface, we used them as *polymeric artificial solid electrolyte interphase* (SEI) coatings on the surface of silicon electrodes. Excellent enhancement in cyclability, rate performance, and interfacial properties was observed, and a focused effort was

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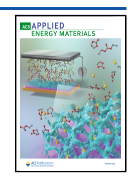

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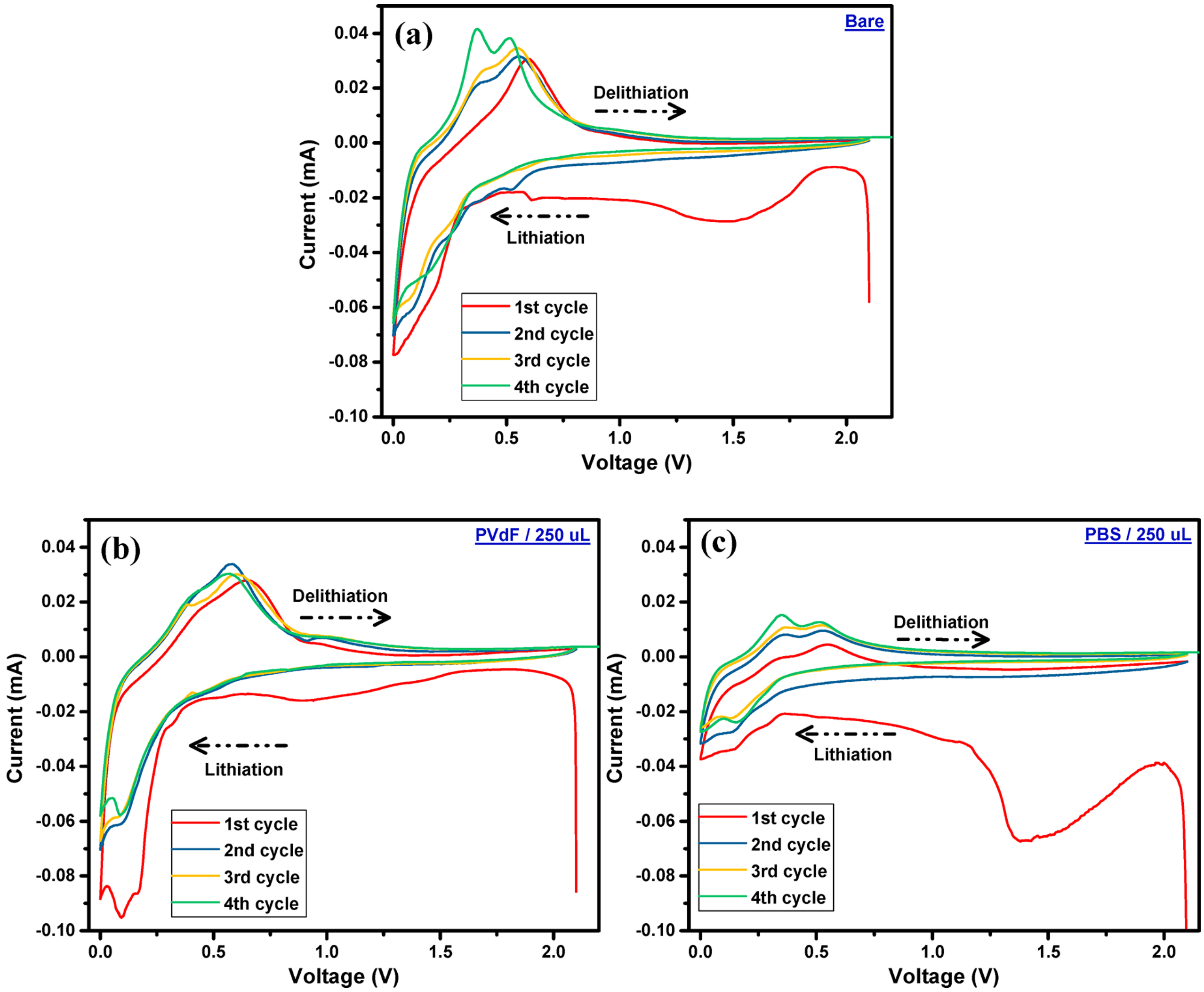

Figure 1. CV studies in anodic half-cells of (a) bare sputter-coated Si anodes, (b) PVdF-coated Si/SiO_2 thin film electrodes, and (c) PBS-coated Si/SiO_2 thin film electrodes.

made to understand the mechanistic origin of enhancement through studies on the defined PBS polymer.

2. EXPERIMENTAL SECTION

Synthesis of PBS. PBS was synthesized through a previously reported procedure and was used as is (Figure S1, Supporting Information). The polymer was characterized by ^{11}B NMR (Figure S2), ^{1}H NMR (Figure S3), and gel permeation chromatography (GPC) studies for molecular weight determination. The formula for the repeating unit is $C_{22}H_{24}BO_2Si$.

Self-Healing Studies. Fifty microliters of a 10^{-2} M solution of PBS in tetrahydrofuran (THF) was prepared and drop-casted on a small rectangular piece of Si-sputtered (~100 nm thickness) copper foil (~15 μm thickness). The electrodes were kindly provided by Nissan Corporation. The solvent was left to evaporate at room temperature and a scar was made artificially to observe self-healing. The scarred polymer-coated electrode was then heated at 45 °C for around 30 min to observe self-healing. FE-SEM images before and after heating were compared to observe self-healing on the electrode surface.

Materials Characterization. ^{1}H and ^{11}B NMR spectra were obtained with a Bruker Advance II-400 MHz spectrometer. Chemical shifts are reported in ppm using the signals corresponding to the residual protons of the indicated deuterated solvent as an internal standard. The molecular weight of the polymer sample was determined at 35C on an LC20AD gel permeation chromatograph (Shimadzu Corporation) equipped with a TSK gel column (G3000H) and a refractive index detector (RID-10A). THF was used as an eluent with a flow rate of 1.0 mL min^{-1}. Calibration was performed with polystyrene standards. SEM images were obtained on an Hitachi S-4500 FESEM instrument.

Electrochemical Measurements. For battery tests, CR-2025 size coin cells were employed by assembling the sputtered silicon anodes mentioned above and Li metal as the counter electrode in an anodic half-cell setup with a polypropylene separator (25 μm, Celguard 2500) and 1 M $LiPF_6$ in EC:DEC (Sigma-Aldrich) as an electrolyte. To see the effect of the polymer, varied amount of PBS solution in THF was drop-casted onto the electrode surface and was left to dry at room temperature. THF was used as a solvent as it is sufficiently polar to dissolve the polymer but at the same time unreactive toward other components. The cells were assembled inside an argon-filled glovebox to avoid moisture contamination (UNICO UN-650F, H_2O and O_2 content <0.1 ppm). The battery charge/discharge tests were performed using a battery cycler (HJ-SD8, Hokuto Denko, Japan) at room temperature. All the other electrochemical techniques described below were performed on a VSP potentiostat (BioLogic) electrochemical analyzer/workstation. Electrochemical impedance spectroscopy (EIS) measurements were performed for the cells after 100 charge−discharge cycles over a frequency range from 100 kHz to 10 mHz with a sinus amplitude of 10 mV. Cyclic voltammetry (CV) was used to determine the electrochemical behavior of the polymer binder at room temperature.

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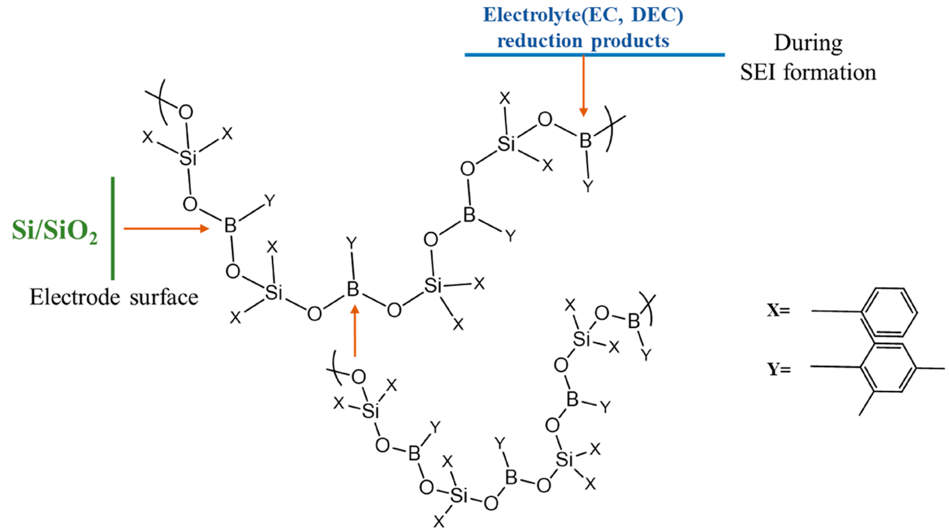

Figure 2. Struture and different possible points of interactions of PBS with other components.

Potential scans were carried out between OCP and 0.0 V versus Li^+/Li at a constant rate of 0.1 mVs^{-1} to determine electrochemical behavior. For charge−discharge measurements, the C-rate was calculated by using the theoretical capacity of Si anodes to be 3579 mA h g^{-1}. For long cycling experiments, the assembled cells were first cycled at 0.1C for five cycles followed by long cycling at 1C. The sputtered Si electrodes were not prelithiated. For full cell studies, the individual electrodes were initially precycled in a half-cell configuration as mentioned, before using them in full cells. For precycling of Si anodes, the coated samples were initially cycled for 1 cycle at 0.1C within the same potential window used for half-cell studies and then lithiated to 0.01 V with respect to (wrt) Li/Li^+, where it was maintained for 60 min. Similar precycling was also performed for $LiMn_{1/3}Ni_{1/3}Co_{1/3}O_2$ (NMC) electrodes (Piotrek, Japan) in a cathodic half-cell setup with constant voltage treatment at 4.3 V wrt Li/Li^+. The full cells were then fabricated with the precycled electrodes and cycled between 3.0 and 4.3 V. The cells were initially cycled for one cycle at 0.1C (wrt Si anode) before performing long cycling studies at 1C (wrt Si anode). The charging protocol consisted of constant current charging plus an additional constant voltage step at 4.3 V for 30 min and then discharge under a constant current mode only. For all full-cell studies, commercial electrolyte (1 M $LiPF_6$ in EC:DEC) with 2 mg/mL each of vinylene carbonate (VC) and fluoroethylene carbonate (FEC) as an additive was used.

Theoretical Studies. Geometry optimization and the electronic properties of the system were evaluated using Gaussian 09 software. B3LYP hybrid functional was employed with 6-311G(d,p) basis set.

3. RESULTS AND DISCUSSION

Cyclic voltammetry (CV) studies were performed to understand the formation of the interfacial film on silicon anodes. Figure 1a shows the CV of the bare sputter-coated Si electrodes. The initial broad peak at ∼1.5 V can be ascribed to the reduction of electrolyte components followed by the onset of lithiation beginning at ∼0.3 V to form amorphous Li−Si alloy at lower potentials. Lithiation results in a core−shell structure having an amorphous outer Li−Si alloy with an inner crystalline silicon domain. The first delithiation process exhibits a single broad peak whereas subsequent cycles show two distinct peaks corresponding to the formation of two different types of amorphous silicon.[16] The ratio of $I_{a.p.}/I_{c.p.}$ (anodic peak current/cathodic peak current) was ∼0.37 (for the second CV cycle), indicating high irreversible lithiation even in the second cycle. Poly(vinylidene fluoride) (PVdF) (commercially utilized binder for Li-ion battery anodes)-coated electrodes showed similar behavior (Figure 1b) to that of bare Si electrodes with the onset of SEI formation at ∼1.5 V but extending over a broad range until 1 V. This can be due to restricted access to the crystalline Si surface because of the ionically insulating PVdF coating.[17,18] This is also reflected in the delithiation cycles where a single broad peak is seen instead of two distinct peaks in the case of the bare electrode after the first cycle. The ratio of $I_{a.p.}/I_{c.p.}$ (anodic peak current/cathodic peak current) in this case was ∼0.22 (for the second CV cycle), indicating further high irreversible lithiation in the second cycle as compared to that of the bare electrode. PBS-coated electrodes showed a broad and intense electrolyte degradation peak around 1.5 V toward SEI formation followed by stepwise lithiation and distinct delithiation peaks from the first cycle onwards (Figure 1c). However, the area integral of the SEI formation peak in the case of the PBS peak was the highest among the three cases, suggesting reductive interaction of electrolyte components with PBS and with the electrode. The boron atom in PBS is well-known to form B−O dative bonds under ambient conditions. Hence, during SEI formation, reduced components of the electrolyte containing oxygen atoms can easily form a cross-linked network with PBS through dynamic B−O dative bonding, which can help in adherence of the electrolyte-derived SEI layer on PBS (Figure 2). This dynamic nature has been explained previously in many other contexts of flexible conductors[19] and can be an important contributor to robust SEI in this case, apart from contributing to the self-healing aspect of the electrochemically modified PBS/SEI layer. The ratio of $I_{a.p.}/I_{c.p.}$ in this case was ∼0.6 (for the second CV cycle), indicating much lower irreversible lithiation even in the second cycle as compared to that of the bare electrode or PVdF-coated electrode. Our group had previously reported the self-healing property of non-cross-linked PBS through interdiffusion of dangling polymer chains via reptation.[15] The self-healing property of PBS on sputter-coated Si/SiO_2 electrodes was evaluated in this case. Since the polymer solution was directly coated on the electrode surface, SiO−B (polymer chain) dative bonds can be expected, which help in adherence of the polymeric structure on the electrode (Figure 2). A mechanical scar on the surface healed almost completely in less than 30 min (heated at 45 °C to expedite the process) (Figure 3), which can be attributed to both

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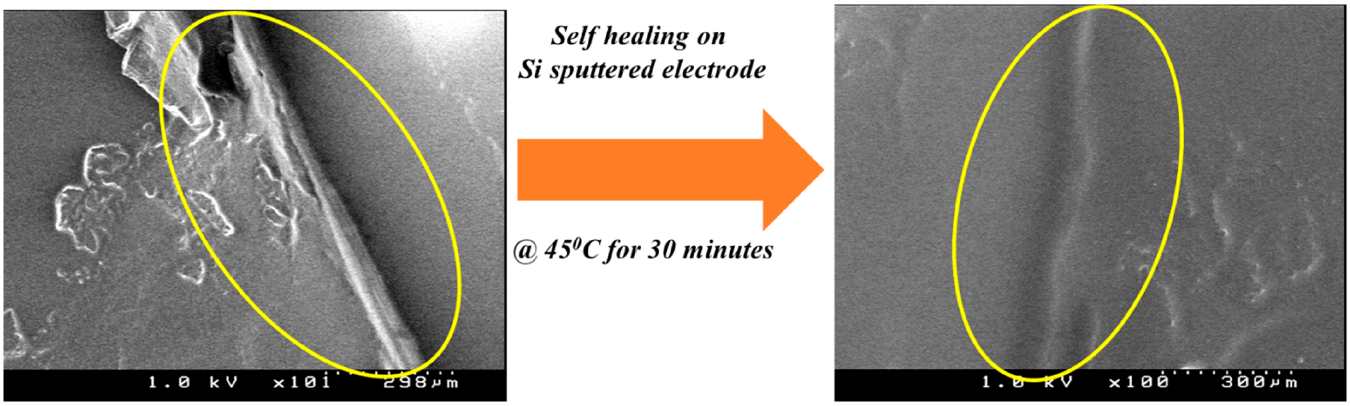

Figure 3. SEM showcasing the self-healing property of PBS on Si/SiO_2-sputtered electrodes.

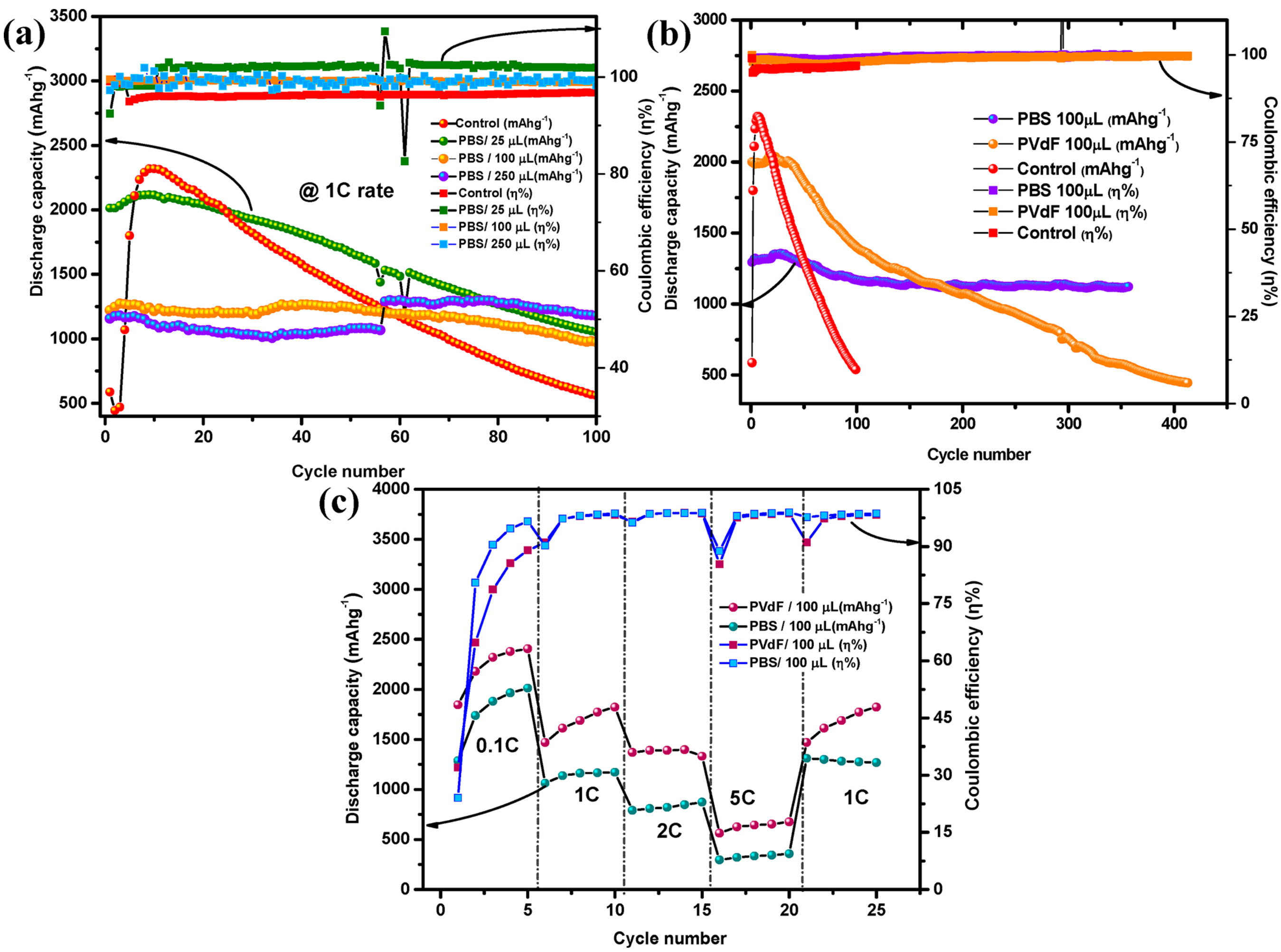

Figure 4. (a) Effect of increasing PBS coating on reversible capacity retention and corresponding Coulombic efficiency. (b) Comparison of cycling stability with PBS- and PVdF-coated samples. (c) Rate performance of PBS- and PVdF-coated samples.

interdiffusion of the PBS chains as well as a contribution from dative bonding.

To understand the effect of PBS coating on the electrode performance, cycling of anodic half-cells was performed with different amounts of PBS-coated electrodes (Figure 4a). A considerable decrease in discharge capacity retention was observed when only 25 μL of PBS solution was coated onto the electrodes. However, when the coating amount was increased, the cycling stability was enhanced with almost 100% reversible capacity retention, even after 100 cycles at 1C in the case of the electrode with 250 μL of the coating. We compared the performance of this optimized coating amount with a similar coating of PVdF and with the bare electrodes toward long cycling stability (Figure 4b, Figures S4 and S5). The bare electrodes showed a rapid reduction of discharge capacity retention upon cycling, with less than 20% of the initial capacity after 100 cycles. In the case of PVdF-coated electrodes, a very high discharge capacity was observed in the

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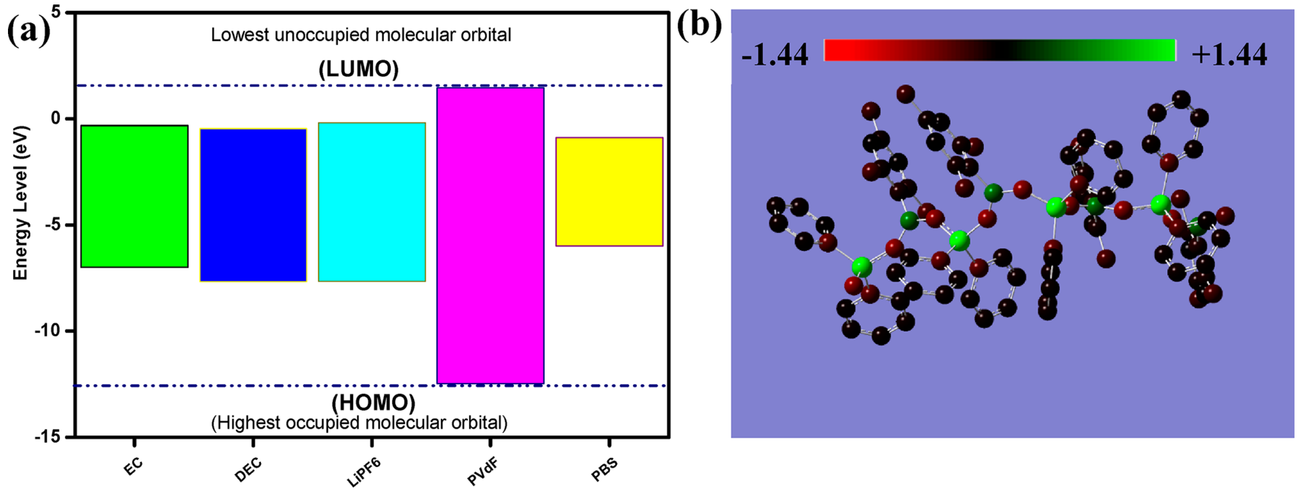

Figure 5. (a) Comparison of energy levels of PVdF and PBS with electrolyte components. (b) Mulliken charge density analysis of optimized polymeric structure (colored as per relative charge density).

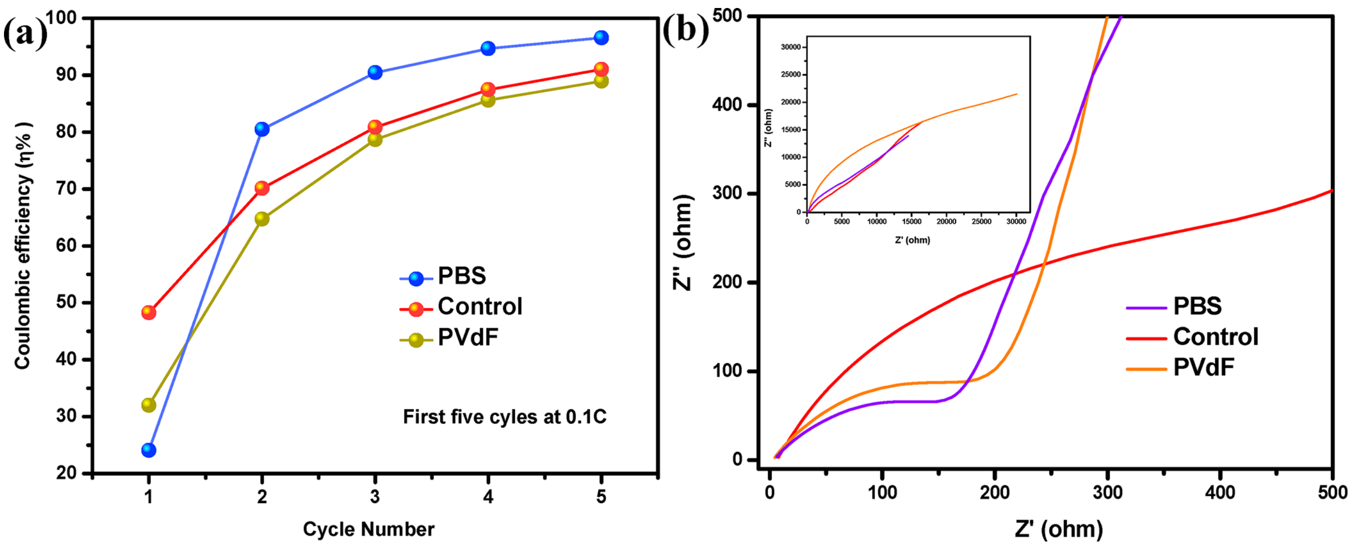

Figure 6. (a) Comparison of Coulombic efficiency during initial cycling for bare Si, PBS- and PVdF-coated electrodes. (b) Comparison of impedance spectra of the various studied electrodes after 100 cycles.

initial cycles followed by a severe drop in capacity. However, PBS-coated electrodes showed a stable reversible capacity retention profile without much loss for over 350 cycles. The rate for PVdF-based coatings was better compared to that of PBS because of the continuous availability of a new silicon surface for lithiation in the short term (detrimental during long cycling, as explained later). PBS-coated electrodes also exhibited excellent rate capability (albeit lower than PVdF) with >300 mA h g^{-1}, even at 5C, while retaining almost similar capacity upon reversal to lower rates (Figure 4c). All this can be explained on the basis of the effect of different polymeric coatings on interfacial properties. The lack of any protective coating in the first case (no polymeric coating) leads to undisrupted volume expansion with related continuous SEI formation with cycling, resulting in a drastic decrease in capacity. Since the active material layer is very thin (~100 nm), it can be assumed that almost all the active material is directly in contact with the electrolyte in the first case, which further aggravates the situation. In the case of PVdF, the polymeric nature helps restrict the volume expansion to some extent, but the Li^+ ion blocking nature of PVdF along with its stiff nature does not help mitigate the issue for long; as a result, drastic volume expansion occurs and provides a new surface for lithiation, which explains the higher rate capacity. But, after the initial 25 cycles, the capacity starts to drop during long cycling studies. Similarly, in the case of PBS-coated electrodes, the entire active material loading can be assumed to be in contact with the protective artificial SEI layer. In this case, the initial attainable discharge capacity was relatively less compared to the former cases. This can be explained by the fact that Li^+ ions are utilized not only by the electrode but also by the components of the polymer because of their low-lying LUMO levels (Figure 5a), which become electrochemically reactive at lower potentials wrt Li, unlike the case of PVdF, which has high-lying LUMO and hence is unreactive in the potential range. However, this excessive irreversible reservoir of Li^+ ions utilized during the initial SEI formation helps create a Li^+ ion conductive polymeric SEI, which is robust and accommodates the volume expansion of Si anodes because of its self-healing capability. This is well-reflected in the Coulombic efficiency (CE) plots of various polymer-coated samples during their initial cycles (Figure 6a). In the case of the bare electrode, the

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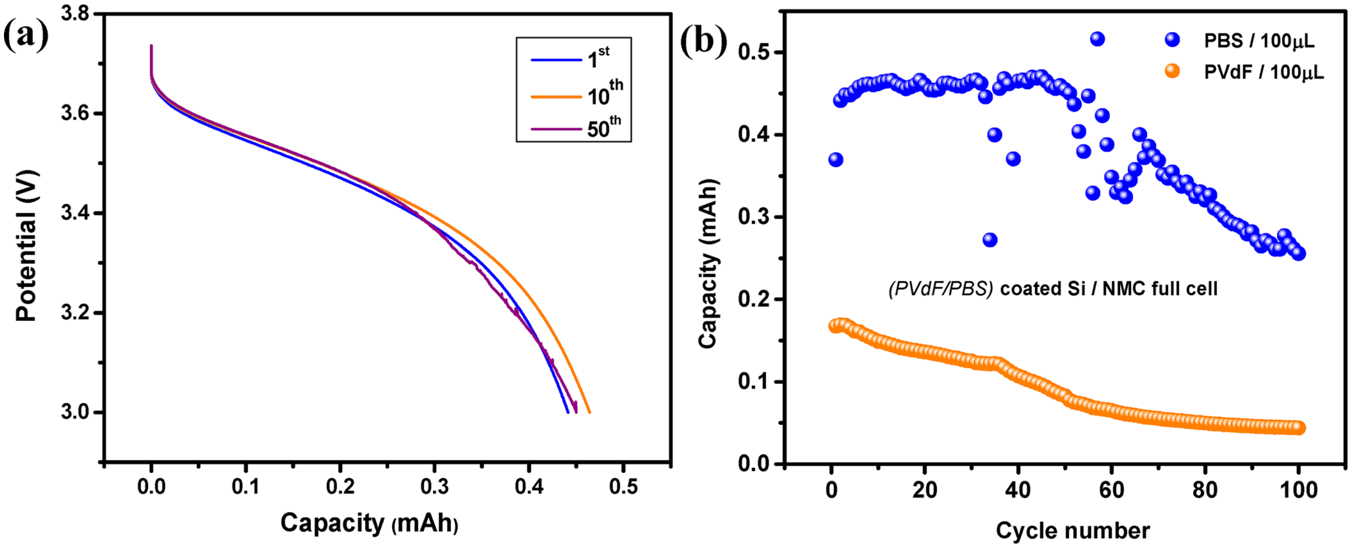

Figure 7. (a) Discharge capacity profiles of PBS-coated electrodes in the full-cell configuration. (b) Comparison of the full-cell cycling stability of polymer (PBS/PVdF)-coated Si anodes at 1C rate.

extent of side reactions was the least and all the electrolyte degradation was due to the interaction of electrolyte components with the active material, resulting in the highest initial CE. But the absence of any volume change buffering agent (like PVdF/PBS) leads to continuous SEI formation in subsequent cycles and hence a drop in CE. PVdF mitigates volume expansion to some extent while continuously degrading electrolyte components on its surface. In the case of PBS, CE increased to >80% after the first cycle because of its Li^+ ion conducting nature and also its self-healing nature, which partially prevents the continuous SEI formation due to volume expansion. The Li^+ conducting nature stems from alternating Si−O linkages as well as from the anion-trapping effect of boron moieties,[20] which has already been established by Matsumi et al. in the past.[21,22] To understand the electronic nature of boron in PBS, Mulliken charge density analysis (Figure 5b) was performed on the optimized conformation of the polymer (tetramer) (Figure S6), which shows the relative charge density on different elements in the polymer chain (Si (+1.445), B(+0.551), and O(−0.563) as their highest charges). The individual relative charges are also tabulated in Table S1. The boron atom is still sufficiently electropositive along with Si alternating with the electronegative O atom, which can help in efficient Li^+ ion conduction through the polymeric network. The interfacial trends concluded from Coulombic efficiency and charge−discharge performance also fall in line with the results of electrochemical impedance spectroscopy (EIS) (Figure 6b). The EIS measurements show the highest interfacial resistance (semicircle in the high-frequency region) in the case of bare electrodes. The resistance was slightly higher in the case of PVdF-coated electrodes as compared to that of PBS-based electrodes, indicating enhanced interfacial kinetics. The full-cell performance of polymer-coated cells was also evaluated with commercial $LiMn_{1/3}Ni_{1/3}Co_{1/3}O_2$ (NMC) cathodes to understand the compatibility of PBS during high-voltage operation (Figure 7a,b). Full cells assembled with PBS-coated Si (Figure 7a) anodes exhibited much higher capacity and cyclability compared to those with PVdF (Figure S7)-coated samples after precycling.

4. CONCLUSION

We report the successful application of defined poly(borosiloxane) as an artificial SEI layer for silicon anodes with enhanced cycling stability and superior rate kinetics. Such an enhancement can be explained by the self-healing nature of the polymer coupled with beneficial contributions from the tricoordinate boron. Computational evidence along with electrochemical characterizations show enhanced interfacial properties of the PBS coating. We also demonstrate full-cell cycling of the coated electrodes, showcasing their applicability in conjugation with high-voltage cathode systems. These findings thus illustrate the potential of defined polymer coatings for high-performance battery electrodes while also opening up their possible scope of application as high-quality self-healing binders for both anodes and cathodes through clever molecular-level engineering of functional groups, which will be evaluated in the future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acsaem.0c02749.

> Figures S1−S7 show the synthetic scheme for PBS, ^{11}B and ^{1}H NMR of PBS, galvanostatic charge−discharge behaviors of PBS- and PVdF-coated electrodes before and after long cycling, optimized structure of PBS (tetramer) with individual atoms numbered, and discharge capacity profiles of PVdF-coated electrodes in the full-cell configuration; Table S1 shows a tabulation of individual charges from Mulliken analysis (PDF)

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Author Contributions

[#]S.G.P. and T.P.J. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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